Physiological Information-Guided Network for Heart Rate Estimation from Near-Infrared Facial Videos

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***Abstract*— Remote Photoplethysmography (rPPG) based on Near-Infrared (NIR) imaging is an emerging non-contact technique for heart rate (HR) monitoring, particularly suitable for low-light environments due to its illumination robustness. However, the accuracy of existing methods remains unsatisfactory for practical use, primarily due to their inadequate capture of subtle variations and complex features inherent in physiological signals across different temporal scales. To address these issues, a Physiological Information-guided Neural Network (PhysioNN) is proposed, in which the Beat-Rhythm joint Attention Mechanism (BRAM) employs beat-based and rhythm-based attention modules to extract multi-scale temporal features from rPPG signals, and the Physiological Feature Guided Regularization (PFGR) integrates physiological domain knowledge through loss constraints to maintain consistency with actual physiological patterns. Furthermore, a multi-scenario dynamic benchmark for NIR-based HR estimation is constructed, named as XDU-SenseTime dataset, comprising 374 videos captured across six scenarios. Experimental results demonstrate that PhysioNN significantly outperforms other methods, achieving an MAE of 0.90 bpm on the public MR-NIRP dataset and 1.86 bpm on the private XDU-SenseTime dataset. In cross-dataset validation, PhysioNN also exhibits strong generalization, with an MAE of 1.80 bpm when trained on MR-NIRP and tested on XDU-SenseTime. Ablation studies validate that PhysioNN establishes an effective framework for learning physiological representations. Thus, the proposed PhysioNN shows considerable potential in enhancing NIR-based non-contact health monitoring capabilities.**

***Index Terms*— Remote Photoplethysmography (rPPG), Heart Rate Estimation, Physiological Information, Near-Infrared.**

I. Introduction

HEART rate (HR) is a fundamental vital sign that serves as a key indicator for assessing physiological health status. Continuous HR monitoring enables health trend analysis, providing significant value for applications such as health assessment and sleep monitoring. Traditional HR measurement methods, such as electrocardiography (ECG) and photoplethysmography (PPG) are already widely used in clinical practice [1]. ECG [2], as a typical electrical method, records cardiac electrical activity through electrodes placed on the skin. PPG [3], as a typical optical technique, estimates HR by detecting light absorption changes caused by blood volume variations, commonly implemented in devices such as fingertip pulse oximeters and smartwatches. However, they require not only skin contact, which may cause irritation or patient discomfort, but also expensive hardware and medical staff for operation. Thus leads to limitations during long-term monitoring, and presents practical constraints under specific circumstances (e.g., for newborns, burn patients, or drivers).

The existing non-contact HR measurement methods can be roughly categorized into WiFi-based [4], Radar-based [5], and Video-based approaches [6]. Among Video-based approaches, Remote Photoplethysmography (rPPG), is an HR monitoring method, which is practical and cost-effective for deployment, particularly well-suited for short-range applications, especially with increasing prevalence of cameras in consumer devices [6]. rPPG can remotely detect pulsatile information caused by heart activity from subtle and invisible color changes in exposed skin [7], which provides a non-contact and convenient way to estimate HR, attracting increasing attention from researchers [8].

rPPG implementations primarily utilize either RGB or Near-infrared (NIR) imaging modalities. RGB-based methods extract rPPG signals by analyzing subtle changes in facial skin color and achieve well performance under normal ambient lighting conditions [9]–[11]. However, RGB light primarily interacts with the reflected light from shallow skin layer, making it susceptible to skin tone difference, lighting changes, and motion artifacts, which leads to unstable rPPG signals. Moreover, RGB cameras are not suitable for night-time or low-light monitoring due to its low visibility in dark conditions. In contrast, NIR can integrate various wavelengths from the infrared spectrum and has greater penetration depth [12], allowing it to capture more stable physiological signals from subcutaneous tissues in low-light conditions, while exhibiting stronger robustness to lighting variations and skin tone differences [13]. Therefore, many researchers start to apply NIR cameras as the equipment for rPPG signal acquisition.

Building upon these imaging principles, researchers have developed various signal processing methods for rPPG-based HR measurement. In previous studies, several signal-analysis models have been proposed for rPPG-based HR measurement. Verkruysse *et al.* first demonstrated HR extraction from RGB facial videos under stable lighting [14]. Jeanne *et al.* used NIR cameras to estimate HR under dynamic lighting conditions [15], and Van Gastel *et al.* demonstrated motion-robust HR estimation using multiple narrow-band NIR wavelengths [16]. Chen *et al.* applied similar methods to NIR videos, showing improved robustness to illumination changes [17]. However, these models are limited by assumptions like skin reflection model and linear combination of noise.

Recently, deep learning models have been widely used in rPPG-based HR estimation tasks and achieved significant progress. Both 2D-CNNs [10] and 3D-CNNs architectures have been explored for rPPG-based HR estimation. Yu *et al.* introduced a 3D-CNN model PhysNet that directly learns spatiotemporal features to reconstruct rPPG signals [9]. Niu *et al.* later proposed the CVD model, which generates multi-scale temporal representations to enhance robustness [18]. More recent efforts have focused on efficient spatiotemporal modeling and improved generalization. Liu *et al.* developed TS-CAN, a dual-branch network with spatial attention, which integrates tensor-shift modules for lightweight learning [11]. Subsequently, Zou *et al.* introduced a transformer-based model RhythmFormer to capture periodic features across multiple time scales [19].

The above methods, including both signal-analysis and deep learning models, can estimate HR to some extent. However, the majority of the existing methods for NIR-rPPG are constrained by the insufficient representation of heartbeat dynamics and temporal rhythm patterns, which restricts their abilities to capture the subtle variations and complexities of HR signals across different temporal scales. In this paper, a robust HR measurement model is proposed, called *PhysioNN* (Physiological Information-guided Neural Network), to effectively capture and analyze physiological signals from NIR facial videos. Furthermore, to evaluate the robustness of NIR-rPPG models in diverse scenarios, we construct a comprehensive benchmark database, called *XDU-SenseTime*, including 374 videos from 17 subjects. This database includes various conditions, establishing less-constrained scenarios of HR estimation.

The main contributions are summarized as follows:

- The PhysioNN model is proposed to improve HR estimation from NIR facial videos by effectively capturing sufficient physiological features across different temporal scales.
- The proposed model enhances robustness by hierarchically integrating beat-based and rhythm-based attention modules to jointly characterize local heartbeat dynamics and global temporal rhythm patterns.
- The proposed model incorporates physiological domain knowledge into training losses to ensure signal alignment in waveform morphology, trend consistency, heartbeat count, interval accuracy, and energy distribution.
- Extensive experimental validation demonstrates the proposed method's superior performance over existing approaches across multi challenging scenarios.

The remainder of this paper is organized as follows. Section II describes the specific architectural details of the proposed PhysioNN model. Section III analyzes the experimental results on two NIR datasets to demonstrate the superior performance of PhysioNN. Finally, we conclude this work in Section IV.

II. METHOD

This section describes the framework of proposed PhysioNN, as illustrated in Fig. 1. It consists of four key components: the *Preprocessing* stage, the *Beat-Rhythm joint Attention Mechanism* (*BRAM*) module, the *Physiological Feature Guided Regularization* (*PFGR*) module, and the *HR Measurement* stage.

A. Preprocessing

In preprocessing, raw data from NIR facial videos are preprocessed (Section II-A). This stage involves detecting and extracting Regions of Interest (ROIs), followed by segmentation of facial signals into fixed-length windows via a sliding window algorithm.

*1) **ROI Detection**:* Our method achieves HR estimation by detecting and analyzing subtle and periodic changes in facial blood flow caused by the cardiac cycle. Typically, these changes are most detectable on specific facial areas, called ROIs, such as the forehead, two cheeks, and chin areas, where blood vessels are highly concentrated and relatively less affected by head movements. Therefore, we detect these ROIs from facial videos to capture and analyze facial signals.

The open-source MediaPipe FaceMesh [20] is applied for facial detection, which provides robust 3D facial landmarks. By projecting these landmarks into 2D coordinates, stable ROI region $\mathcal{R}$ can be located and updated frame by frame, even in the presence of slight head movements. This helps reduce motion-induced noise and ensures consistent signal extraction across varying poses.

*2) **rPPG Extraction**:* In remote HR measurement, preprocessing the color intensity signal derived from facial ROIs (detrending, normalization, filtering, and denoising) improves signal quality and reduces the impact of noise. For each frame I_t, the raw rPPG signal is extracted by averaging the pixel intensities within the facial ROI region R as: $s(t) = \frac{1}{|R|}\sum_{p\in R} I_t(p)$, where R denotes the set of pixels inside the selected ROI, $|R|$ is the number of pixels in R, p indexes a pixel within the ROI, and $I_t(p)$ denotes the intensity value of pixel p in frame I_t.

Given the raw signal $s(t)$, the specific process is as follows:

- Detrend the input signal: $s_d(t) = s(t) - p(t)$, where $s_d(t)$ is the detrended signal, $p(t)$ is the fitted polynomial. This can remove baseline drift as well as some linear or nonlinear trends within the signal.
- Normalize the signal using Z-score: $s_z(t) = \frac{s_d(t)-\mu}{\sigma}$, where $s_z(t)$ is the normalized signal, where μ and σ are of the mean and standard deviation of $s_d(t)$, respectively. This transforms the data into a standard normal distribution, thus remove the effects caused by factors such as lighting variations, noise, and motion artifacts.
- Filter the signal using Band-Pass filter: $s_f(t) = Filter(s_z(t), f_{low}, f_{high})$, where $s_f(t)$ is the filtered signal, $Filter(\cdot)$ represents the Butterworth Filter, f_{low} and f_{high} represents the corresponding lower and upper cutoff frequencies of the filter. In this paper, we use a frequency range of 0.75 Hz to 3 Hz, corresponding to a HR of 45 to 180 bpm.
- Denoise the signal using wavelet transform: $s_n(t) = \mathcal{W}^{-1}(\mathcal{T}(\mathcal{W}(s_f(t))))$, where $s_n(t)$ is the denoised signal, $\mathcal{W}$ and $\mathcal{W}^{-1}$ represent the forward and inverse

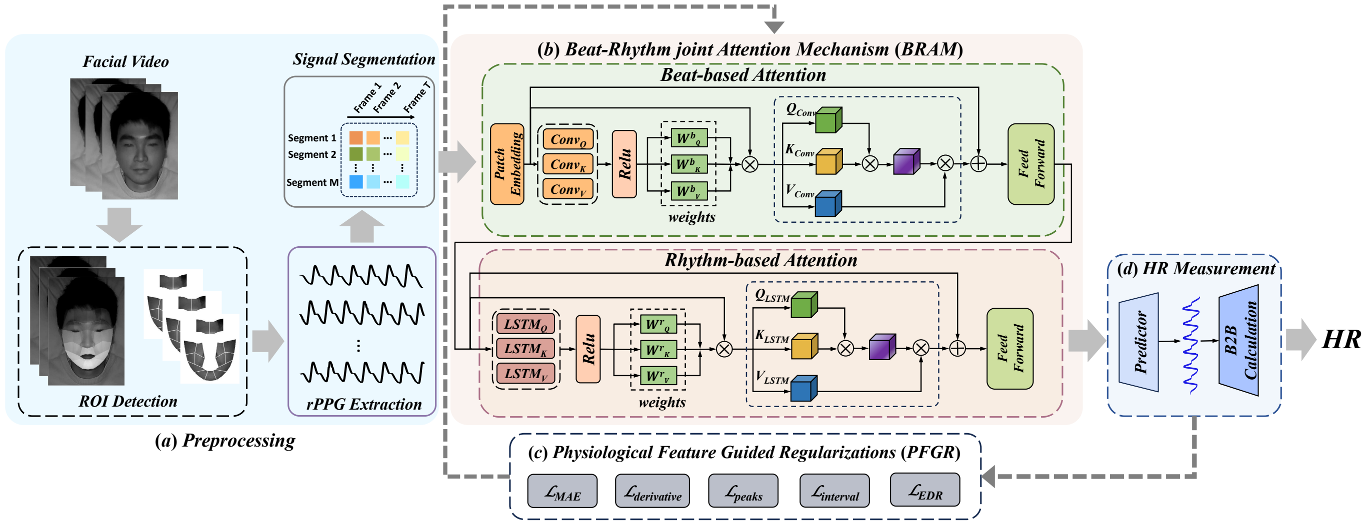

Fig. 1. Overview of the PhysioNN framework. (a) The *Preprocessing* stage. (b) The *BRAM* module. (c) The *PFGR* module. (d) The *HR Measurement* stage.

wavelet transform operations respectively, $\mathcal{T}(\cdot)$ denotes the thresholding function applied to the wavelet coefficients. The wavelet transform is employed to further denoise the signal, producing a clean, noise-free rPPG signal.

3) Signal Segmentation: The resulted rPPG signal $s_n(t)$ has a total length of L. The signal is divided using a fixed window size T into M consecutive segments, *i.e.*, $L = T \times M$. Let

$$x = \{x^{(1)}, x^{(2)}, \dots, x^{(m)}, \dots, x^{(M)}\} \tag{1}$$

denote the set of segments, $x^{(m)} \in \mathbb{R}^T$ being the m-th window of length T extracted from $s_n(t)$. The set x is then fed into the BRAM Module for feature learning.

B. Beat-Rhythm joint Attention Mechanism (BRAM)

The processed signal segments are fed into the BRAM module to capture multi-scale temporal features of heartbeat signals.

1) Beat-based Convolutional Attention: As the first part of BRAM, the Beat-based Convolutional Attention in Fig. 1 (b) employs one-dimensional convolution and self-attention to adaptively capture and enhance key temporal features within the heart rate signal, such as heartbeat peaks and RRI (R-R intervals, the time difference between successive R waves).

Initially, it applies a differentiation operation, $diff(\cdot)$, to the input sequence x to capture its dynamic characteristics and short-term variations. Subsequently, three separate convolutional layers are applied to the differentiated signal to generate the normalized weight matrices W_Q^b, W_K^b, and W_V^b, which are then used to project the original input signal x to form the query (Q_{conv}), key (K_{conv}), and value (V_{conv}) features. This process can be conveniently represented using the following equations:

$$\begin{aligned} Q_{conv} = x \cdot W_Q^b &= x \cdot norm(Conv_Q(diff(x)))^T, \\ K_{conv} = x \cdot W_K^b &= x \cdot norm(Conv_K(diff(x)))^T, \\ V_{conv} = x \cdot W_V^b &= x \cdot norm(Conv_V(diff(x)))^T, \end{aligned} \tag{2}$$

where $norm(\cdot)$ denotes normalization, and $Conv_Q$, $Conv_K$, $Conv_V$ are one-dimensional convolutions capable of extracting diverse aspects of the heartbeat rhythm.

The self-attention mechanism is then adopted to compute a refined representation by weighting the value features with the compatibility between query and key features. The final output x_b is produced through a residual connection:

$$\begin{aligned} x_b &= x + Attention_{Beat} \\ &= x + \text{softmax}\left(\frac{Q_{conv} K_{conv}^T}{\sqrt{d_b}}\right) V_{conv}, \end{aligned} \tag{3}$$

where d_b is the dimension of the query features Q_{conv}.

The convolutional operations are essential for extracting localized temporal patterns (e.g., the upstroke and downstroke of individual heartbeats), while the attention mechanism focuses on the key heartbeat-related features, such as peak locations and RRI that are important for HR estimation. By filtering out irrelevant components and noise, the Beat-based Convolutional Attention module enhances the model's capability to extract meaningful physiological features, enhancing its performance in accurately modeling HR dynamics, even in the presence of subtle lighting variations.

2) Rhythm-based Recurrent Attention: The Rhythm-based Recurrent Attention is designed to capture the periodic trends and long-term dependencies of heartbeat signals.

An LSTM network is utilized to extract temporal features from x_b, as LSTM is effective at learning dependencies over time and is well-suited for handling the sequential nature of heartbeat signals. The signal x_b is multiplied by the normalized weights W_Q, W_K, and W_V through matrix operations to generate the Query (Q_{LSTM}), Key (K_{LSTM}), and Value (V_{LSTM}).

$$\begin{aligned} Q_{LSTM} &= x_b \cdot W_Q^r = x_b \cdot norm\left(Linear_Q\left(LSTM_Q\left(x_b\right)\right)\right), \\ K_{LSTM} &= x_b \cdot W_K^r = x_b \cdot norm\left(Linear_K\left(LSTM_K\left(x_b\right)\right)\right), \\ V_{LSTM} &= x_b \cdot W_V^r = x_b \cdot norm\left(Linear_V\left(LSTM_V\left(x_b\right)\right)\right), \end{aligned} \tag{4}$$

where $Linear_Q$, $Linear_K$, and $Linear_V$ are three linear layers.

The rhythm-based attention mechanism then computes a refined representation. And the final output x_r is also produced through a residual connection:

$$\begin{aligned} x_r &= x_b + Attention_{Rhythm} \\ &= x_b + \text{softmax}\left(\frac{Q_{LSTM}K_{LSTM}^T}{\sqrt{d_r}}\right)V_{LSTM}. \end{aligned} \tag{5}$$

where d_r is the dimension of the query features Q_{LSTM}.

Overall, the rhythm-based attention module enhances the model's ability to focus on long-term rhythmic patterns, such as sustained heartbeat trends and periodical variations. By capturing these global trend features, it improves the model's ability to accurately estimate long-term HR dynamics.

C. Physiological Feature Guided Regularization (PFGR)

The BRAM output signals are constrained by PFGR's regularization terms to align with physiological characteristics of actual heartbeat signals. The structure of the PFGR is shown in Fig. 1 (c). Incorporating domain-specific physiological characteristics into loss function enables the model maintain consistency with real heartbeat signals, to enhance the accuracy and physiological plausibility of the output signals. Studies have shown that the cardiac cycle-related features can be categorized into the following three types [21].

1) Heartbeat Dynamics: Heartbeat dynamics refer to the variation patterns of heartbeat signal. These dynamics can be captured through features such as the signal's amplitudes (overall waveform) and instant fluctuations, which can be measured via two loss functions $\mathcal{L}_{\text{MAE}}$ and $\mathcal{L}_{\text{derivative}}$, respectively.

$\mathcal{L}_{\text{MAE}}$ measures the overall difference between the predicted and actual BVP (Blood Volume Pulse) signals, reflecting the accuracy of the overall signal. It can capture the global heartbeat dynamics by focusing on the general shape of the output rPPG signals. $\mathcal{L}_{\text{derivative}}$ measures the difference within heartbeat fluctuations over time. It captures the dynamic variations of the rPPG signal, emphasizing temporal changes in the heartbeat dynamics.

$$\mathcal{L}_{\text{MAE}} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|, \tag{6}$$

$$\mathcal{L}_{\text{derivative}} = \frac{1}{N-1}\sum_{i=1}^{N-1}|\Delta y_i - \Delta\hat{y}_i|, \tag{7}$$

where N denotes the number of samples. y_i and $\hat{y}_i$ are the ground-truth PPG signals and the predicted rPPG signals of the i-th subject; Δy_i and $\Delta\hat{y}_i$ are the ground-truth and the predicted changing rate of the i-th subject, respectively.

2) Pulse Wave Morphology: Pulse Wave Morphology includes shape and characteristics of pulse wave, such as peak positions, time intervals, and the rise and fall rates of waveform.

$\mathcal{L}_{\text{peaks}}$ and $\mathcal{L}_{\text{interval}}$ represents heartbeat count regularization and heartbeat interval regularization. These two regularization terms ensure that the heartbeat positions of the predicted signal align with those of the real BVP signals, to enable the model to capture subtle temporal dynamic changes and adapt to individual diversity.

$$\mathcal{L}_{\text{peaks}} = |N_{\text{pred}} - N_{\text{target}}|, \tag{8}$$

$$\mathcal{L}_{\text{interval}} = mean\left(|\text{diff}(N_{\text{pred}}) - \text{diff}(N_{\text{target}})|\right), \tag{9}$$

where N_{pred} and N_{target} refer to the lists of peak positions in the predicted and ground-truth signals, respectively. $\text{diff}(\cdot)$ represents time differences between consecutive elements in the lists and $mean(\cdot)$ denotes the arithmetic mean operator.

3) Temporal Rhythm Patterns: Temporal rhythm patterns capture dynamic changes and rhythmic variations of the rPPG signal in time domain, such as energy distribution in a fixed time interval.

The energy distribution regularization $\mathcal{L}_{\text{EDR}}$ ensures the output signal aligns with the energy distribution of the ground-truth BVP signal. It allows the model to focus on the periodicity and energy fluctuations, while minimizing the impact of noise on energy distribution.

$$E_k = \sum_{t=k}^{k+T} x_t^2, \tag{10}$$

$$\mathcal{L}_{\text{EDR}} = mean\left(\sum_{t=k}^{k+T}|\hat{E}_k - E_k|\right) \tag{11}$$

where k represents the starting position of the current time window, T denotes the sampling length, x_t represents the signal at time t. E_k and $\hat{E}_k$ represent the energy distributions of the ground-truth PPG signal and the predicted rPPG signal within the time interval T, respectively.

Overall, our model is trained using a summed loss function, formulated as follows:

$$\mathcal{L} = \alpha(\mathcal{L}_{\text{MAE}} + \mathcal{L}_{\text{derivative}}) + \beta(\mathcal{L}_{\text{peaks}} + \mathcal{L}_{\text{interval}} + \mathcal{L}_{\text{EDR}}), \tag{12}$$

where α and β are weighting coefficients to adjust the relative importance of each loss component.

D. HR Measurement

In the *HR Measurement* stage, the output of the BRAM module is firstly fed into the *Predictor* to generate the one-dimensional rPPG signal. The Predictor employs a Conv1D layer to capture local dependencies and a Linear layer to adjust the output dimension. Subsequently, Beat-to-Beat (B2B) method [22] is applied to calculate HR values by directly detecting RRI, as shown in Fig. 2.

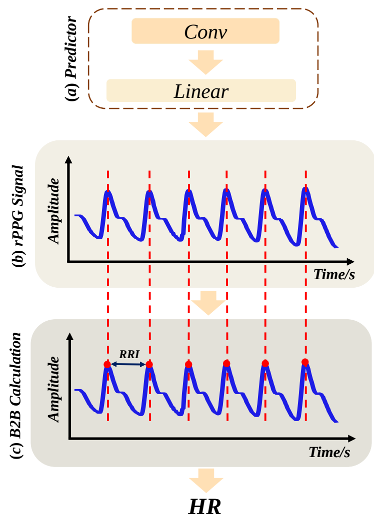

Fig. 2. The illustration of the *HR Measurement*.

By obtaining HR information from each individual heartbeat interval, the B2B method can capture real-time HR changes in less-constrained scenarios. The formula for B2B HR calculation is as follows:

$$HR = \frac{T}{mean(p^{(2)} - p^{(1)}, \cdots, p^{(K)} - p^{(K-1)})}, \tag{13}$$

where T is the window length, $p^{(i)}$ is the position of the $i-$th heartbeat, and K is the total number of heartbeats detected within the current window.

III. Experimental Evaluation

A. Datasets and experimental setups

1) Datasets: Evaluations are conducted on one public dataset MR-NIRP [23] and one private dataset XDU-SenseTime.

MR-NIRP includes RGB and NIR videos from 8 subjects with various skin tones under static and dynamic lighting conditions. The duration of the videos ranges from 1 to 3 minutes, recorded at 30 fps (frames per second) with a resolution of 640×640. **XDU-SenseTime** includes data from 17 subjects, aged between 20 to 40, with participants not wearing masks or glasses. There are 6 different scenarios in XDU-SenseTime dataset, including steady, single side, small rotation, large rotation, talking, post-exercise. The example video frames of one subject captured by different conditions and devices are shown in Fig. 3. The data collection environment is controlled at a temperature of $24 \pm 3°C$ with subjects positioned at a fixed distance of 1 meter from the cameras. Videos were captured in both well-lit and dark indoor scenes using RGB and NIR cameras developed by Xi'an SenseTime Technology, with a resolution of 1280×720, a frame rate of 24 fps, and stored in mp4 format.

2) Experimental setup: The proposed method is implemented under the PyTorch framework with an NVIDIA GeForce RTX 4090 GPU. The video and corresponding PPG signals are sampled using a window of 150 frames with a step size of 30 frames. The parameters α and β in Eq. (12) are set to 0.7 and 0.1 respectively. The model is trained with Adam optimizer and the learning rate and Dropout are 0.001 and 0.11, respectively.

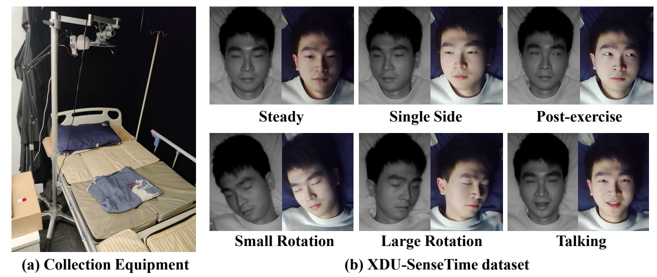

Fig. 3. Details of XDU-SenseTime dataset. (a) The recording setup of XDU-SenseTime dataset. (b) Examples of video frames.

3) Evaluation Metrics: The Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and Pearson Correlation Coefficient (r) are utilized to evaluate algorithm performance.

Moreover, three time-domain indicators are applied to measure HR variability: the mean value (MeanNN), the standard deviation (SDNN), and the root mean square of successive differences (rMSSD).

B. Performance Evaluation

1) Intra-Dataset Testing: Firstly, we conduct intra-dataset testing on all the datasets used in this paper. The proposed PhysioNN is compared with other non-contact methods, including signal-analysis methods (GREEN [14], POS [24], and ICA [25],) and deep learning models (PhysNet [9], DeepPhys [10], TS-CAN [11], RhythmFormer [19], EfficientPhys [26], and IENet [27]).

Performance on MR-NIRP Dataset. Table I shows the comparison results of different methods on MR-NIRP dataset. As shown in Table I, traditional methods such as GREEN and POS exhibit inferior performance. This may be attributed to their reliance on manually designed features, which are insufficient to comprehensively capture complex patterns in HR signals. Among deep learning methods, models exhibit varying performance levels. Spatio-temporal networks (PhysNet, DeepPhys) show sensitivity to motion and illumination artifacts, while attention-based methods (TS-CAN, RhythmFormer) demonstrate improved but still limited performance due to their focus on specific temporal or spatial features without comprehensive physiological constraints. IENet, though incorporating multimodal fusion, struggles with noise suppression under challenging conditions.

Among all methods, the proposed PhysioNN achieves the most competitive results, with 0.90 bpm in MAE, 1.45 bpm in RMSE, and 0.97 in Pearson Correlation Coefficient. The two-level attention mechanism BRAM enables progressively focus on edges and sequential features of physiological signals, reducing the impact of ambient light and noise. Additionally, PFGR incorporates domain-specific physiological constraints to maintain consistency with actual heartbeat patterns.

Performance on XDU-SenseTime Dataset. Table II compares models on the XDU-SenseTime dataset. PhysioNN maintains superior performance on the more challenging

TABLE I
PERFORMANCE COMPARISON ON THE MR-NIRP DATASET.

Method	MAE ↓ (bpm)	RMSE ↓ (bpm)	r ↑
signal-analysis models			
GREEN [17]	3.15	6.02	0.79
POS [28]	4.37	7.95	0.85
deep learning models			
DeepPhys [10]	7.78	16.8	-0.03
PhysNet [9]	3.07	7.55	0.66
TS-CAN [29]	2.52	4.32	0.71
RhythmFormer [30]	3.86	8.64	0.76
IENet [27]	3.62	8.10	0.78
PhysioNN(Ours)	**0.90**	**1.45**	**0.97**

XDU-SenseTime dataset, which includes dynamic scenarios such as head rotations, talking, and post-exercise conditions. While traditional methods fail to model complex noise patterns under motion and illumination variations, deep learning methods exhibit varied adaptability. PhysNet, as a spatio-temporal network, shows limited capacity in capturing fine-grained temporal periodicities, particularly under motion artifacts and illumination changes. DeepPhys achieves relatively low MAE on this dataset but demonstrates significant instability across different scenarios. In comparison, attention-based methods demonstrate more stable performance: TS-CAN utilizes spatial attention to focus on physiologically relevant pixels, thereby reducing noise amplification and achieving stronger performance. RhythmFormer employs hierarchical temporal attention to model periodic patterns, though its performance remains constrained by insufficient physiological constraints. However, these methods still face challenges in maintaining consistency under high-motion scenarios such as talking and large head rotations.

PhysioNN achieves robust performance through its hierarchical attention mechanism and physiological regularization, which collectively suppress motion-induced noise while preserving essential pulse wave characteristics. The slight performance difference compared to MR-NIRP (MAE: 1.86 vs. 0.90 BPM) is attributed to the increased complexity of XDU-SenseTime, yet PhysioNN consistently outperforms all benchmarks, demonstrating its generalization capability across diverse scenarios.

TABLE II
PERFORMANCE COMPARISON ON THE XDU-SENSETIME DATASET.

Methods	MAE ↓ (bpm)	RMSE ↓ (bpm)	r ↑
signal-analysis models			
GREEN [14]	5.74	7.28	0.77
POS [24]	4.15	5.91	0.82
deep learning models			
DeepPhys [10]	2.71	6.08	0.51
PhysNet [9]	4.21	6.36	0.88
TS-CAN [11]	2.40	5.19	0.95
RhythmFormer [19]	3.56	4.78	0.82
IENet [19]	3.98	7.64	0.85
PhysioNN(Ours)	**1.86**	**2.47**	**0.98**

Performance in Different Scenarios. As shown in Table III, the "Steady" and "Post-exercise" scenarios yield optimal performance, primarily attributed to minimal motion artifacts enabling cleaner physiological signals. "Small Rotation" achieves better results than "Large Rotation", as vigorous motion may introduce stronger interference with signal estimation. The "Talking" scenario presents the most significant challenges, where facial muscle activity and subtle head movements severely interfere with physiological signal measurement, resulting in the highest estimation errors among all scenarios. Furthermore, Table III indicates that NIR videos consistently achieve better performance under "Dim" versus "Bright" illumination across all scenarios. This is attributed to the superior performance of NIR cameras in low-light environments, where controlled NIR illumination suppresses ambient visible light interference, yielding cleaner and more stable physiological signals.

TABLE III
EXPERIMENTAL RESULTS ON NIR VIDEOS IN DIFFERENT SCENARIOS.

Scenarios	Illumination	MAE ↓ (bpm)	RMSE ↓ (bpm)	r ↑
Steady	Bright	1.68	2.81	0.96
	Dim	**1.52**	**2.15**	**0.98**
Single Side	Bright	1.75	2.94	0.94
	Dim	1.61	2.38	0.96
Small Rotation	Bright	2.13	3.25	0.92
	Dim	1.85	2.42	0.94
Large Rotation	Bright	2.94	3.75	0.89
	Dim	2.75	3.61	0.91
Post-exercise	Bright	**1.65**	**2.32**	**0.98**
	Dim	1.55	2.20	0.97
Talking	Bright	2.45	3.96	0.93
	Dim	2.31	3.75	0.92

2) Cross-Dataset Testing: In addition to intra-dataset testing, we also perform cross-dataset testing to demonstrate the generalization ability of the proposed method. Specifically, the model trained on one dataset is directly validated on another dataset. For example, MR-NIRP $\rightarrow$ XDU-SenseTime indicates that the model is trained on MR-NIRP and tested on XDU-SenseTime. The parameter settings for the cross-dataset experiments are the same as those used in the original intra-dataset experiments. Table IV presents the comparison results with other existing methods. The results demonstrate that our proposed PhysioNN significantly outperforms others in cross-dataset testing, demonstrating its strong generalization ability in various scenarios.

3) Robust HR Estimation: Twelve challenging scenarios from the MR-NIRP and XDU-SenseTime datasets were selected to evaluate PhysioNN's robustness, as shown in Fig. 4. In the "Still" scenario, the predicted signals demonstrated optimal alignment with ground truth. In "Bright" and "Talk" scenarios, although the rPPG signals exhibited some inconsistencies, their peaks, troughs, and periodic patterns remained well-aligned. Since PhysioNN employs the B2B method for HR estimation, which calculates heart rate from average beat intervals, the predicted signals need only follow the overall trend of the ground truth PPG, rather than precisely matching instantaneous waveform details

In "Beard Occlusion" scenarios, facial hair partially ob-

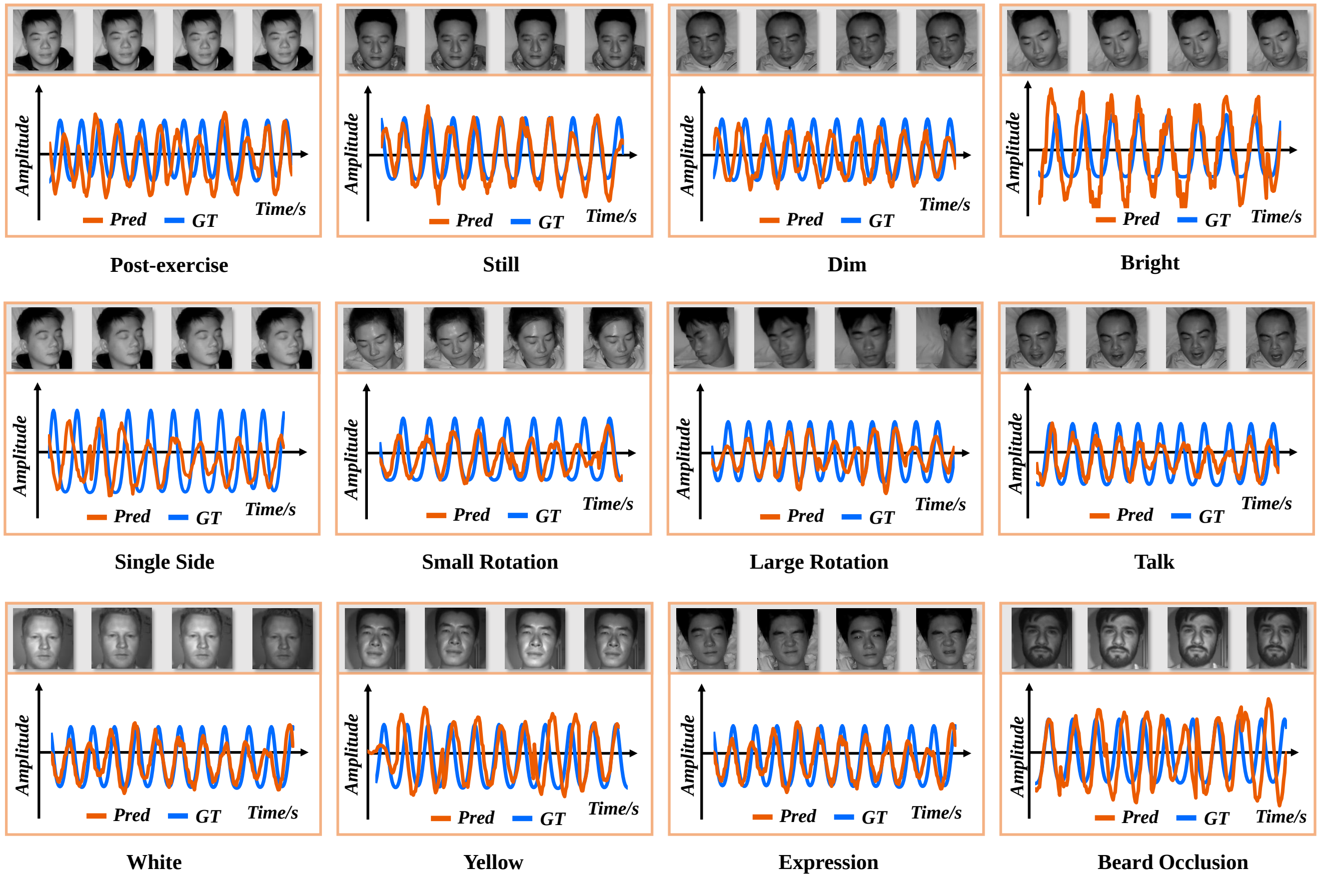

Fig. 4. Visualization of quantitative results across 12 challenging scenarios from two datasets (MR-NIRP and XDU-SenseTime). (Pred: predicted rPPG; GT: ground-truth PPG. Representative video frames are shown above each corresponding signal plot.)

TABLE IV
EXPERIMENTAL RESULTS ON CROSS-DATASET TESTING.

Methods	MR-NIRP → XDU-SenseTime			XDU-SenseTime → MR-NIRP		
	MAE	RMSE	r	MAE	RMSE	r
signal-analysis models						
GREEN [14]	5.15	6.72	0.85	5.42	6.59	0.87
POS [24]	4.28	5.81	0.82	6.38	7.46	0.85
ICA [25]	4.86	5.94	0.84	5.69	6.82	0.87
deep learning models						
DeepPhys [10]	3.42	4.58	0.92	3.57	4.42	0.95
EfficientPhys [26]	4.78	6.13	0.87	7.48	9.72	0.86
PhysNet [9]	2.70	3.58	0.82	4.68	5.32	0.82
TS-CAN [11]	5.82	6.82	0.85	6.78	8.51	0.86
RhythmFormer [19]	6.78	8.41	0.72	9.44	11.32	0.74
PhysioNN(Ours)	**1.80**	**1.85**	**0.97**	**2.89**	**2.49**	**0.98**

structed ROIs, increasing difficulty in spatial feature extraction and causing slight waveform misalignment. However, signal trends remained aligned with ground truth, demonstrating PhysioNN's ability to capture key physiological patterns despite occlusions.

To assess skin tone effects, subjects with light ("White") and medium ("Yellow") skin tones were compared. Lighter tones tended to yield cleaner, more consistent signals, while medium tones introduced marginally more noise and distortion. This suggests melanin concentration influences signal-to-noise ratio under NIR imaging. Nonetheless, predicted signals maintained alignment with ground truth in both cases, highlighting cross-skin-tone robustness.

Crucially, PhysioNN maintained stable HR estimation under significant noise, such as in "Single Side" and "Large Rotation", demonstrating its ability to suppress interference and reliably extract physiological information.

C. Ablation Studies

This section introduces the ablation study results of the proposed method on the MR-NIRP and XDU-SenseTime datasets to analyze the impact of each component on the experimental performance.

1) BRAM: To validate the effectiveness of BRAM module, the network is trained using three models: (1) the Baseline (Transformer), (2) Baseline + Beat-based Attention Module, (3) Baseline + BRAM (Beat-Rhythm joint Attention Mechanism). The comparison results are displayed in Fig.5.

In the time domain, as shown in Fig.5 (a), we present comparison results of the pulse waveform between the predicted rPPG signals and the ground truth PPG signal. It can be observed that the rPPG signal predicted by the original baseline method fail to align with the real signal in terms of peak, valley and the overall signal trend. After incorporating the Beat-based

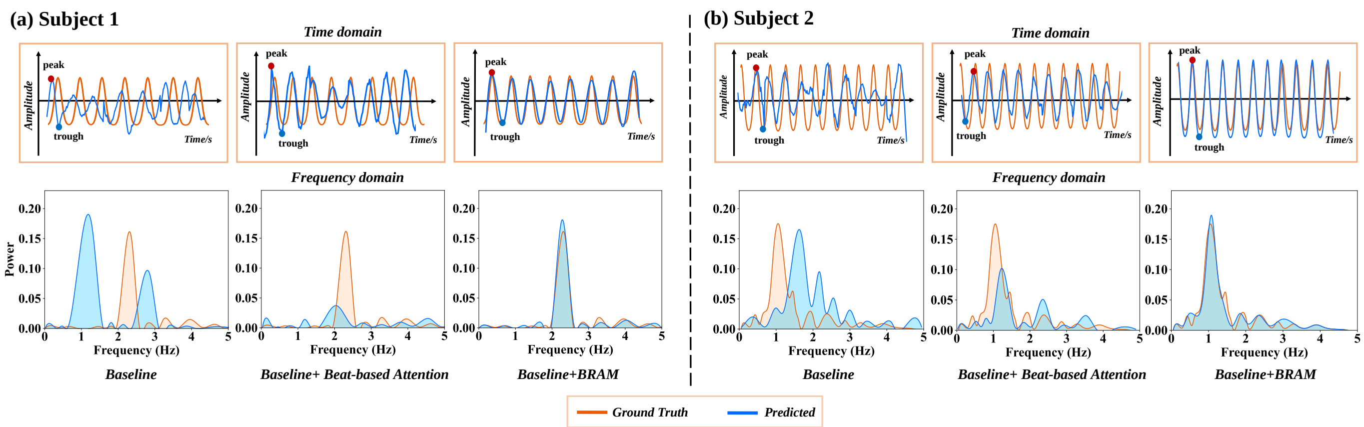

Fig. 5. Ablation experimental results of HR estimation on XDU-SenseTime datasets. The blue and orange lines represent the predicted rPPG signals and the ground-truth PPG signals, respectively. (a) The subject sample 1. (b) The subject sample 2.

Attention module, the predicted rPPG signal shows improved alignment with detailed waveform characteristics of the ground truth signal. By applying the full BRAM module (integrating both beat and rhythm attention), the Baseline+BRAM model not only accurately locates peaks and troughs but also better fits the overall trend of the ground truth signal.

In the frequency domain, as shown in Fig.5 (b), we present a comparison of the power spectral density (PSD) between the predicted rPPG signals and the ground truth PPG signal. The PSD can illustrate the distribution of signal power across different frequencies. It is evident that the baseline method exhibits a poor fit in the frequency domain and fails to capture the signal's main frequency components. The Baseline + Beat-based Attention shows slight improvement, with the frequency characteristics of the signal enhanced to some extent. The Baseline + BRAM module performs the best, with the highest degree of fit in the PSD, accurately capturing both the frequency components and the distribution of the signal. This further proves that rhythm-based attention module can capture the overall trend and frequency distribution of the PPG signal in the entire temporal sequence.

Moreover, the quantitative comparison results are also proposed in Table V. It can be observed that the Baseline shows a 6.36 bpm higher MAE compared to the proposed Baseline + BRAM. This primarily because BRAM can extract multi-scale temporal features from rPPG signals, resulting in better fit for the real signal, thereby improving HR estimation performance.

TABLE V
ABLATION EXPERIMENTAL RESULTS OF VARIOUS SCENARIOS ON XDU-SENSETIME DATASETS.

Baseline	BRAM	PFGR	MAE	RMSE
✓			8.32	9.47
✓	✓		1.96	2.78
✓		✓	7.96	9.13
✓	✓	✓	1.86	2.59

2) PFGR: To validate the effectiveness of PFGR module, the proposed model is compared with the Baseline model, and the results are presented in Table V. After applying PFGR, the HR MAE decreased from 8.32 bpm to 7.96 bpm. This improvement is primarily due to PFGR's heartbeat count and heartbeat interval regularization, which further constrain the model from overfitting to abnormal signals, ensuring that the estimation results align with actual physiological patterns.

D. Parameter Analysis

1) Analysis of Multiple Losses: To investigate the contribution of each loss component in PFGR, we progressively remove individual loss terms during training on both XDU-SenseTime and MR-NIRP datasets. The results in Table VI show that removing any loss component $\mathcal{L}_{\text{MAE}}$ and $\mathcal{L}_{\text{derivative}}$, $\mathcal{L}_{\text{peaks}}$ and $\mathcal{L}_{\text{interval}}$, or $\mathcal{L}_{\text{EDR}}$ consistently degrades performance across all metrics. And optimal performance is achieved when all losses are incorporated. This indicates that the combined loss function is crucial for enhancing HR estimation accuracy, as each component constrains corresponding distinct physiological characteristics.

TABLE VI
ABLATION EXPERIMENTAL RESULTS OF DIFFERENT COMBINATION STRATEGIES OF VARIOUS LOSSES.

Setting	XDU-SenseTime			MR-NIRP		
	MAE	RMSE	r	MAE	RMSE	r
w/o $\mathcal{L}_{\text{MAE}}$, $\mathcal{L}_{\text{derivative}}$	2.88	4.17	0.92	4.38	5.99	0.92
w/o $\mathcal{L}_{\text{peaks}}$,$\mathcal{L}_{\text{interval}}$	2.47	3.61	0.94	4.17	5.48	0.95
w/o $\mathcal{L}_{\text{EDR}}$	1.96	2.67	0.96	0.97	1.38	0.95
all	0.90	1.45	0.97	1.86	2.47	0.98

2) Hyperparameters α and β: To evaluate the impact of hyperparameters α and β, we conduct a quantitative analysis with results shown in Fig. 6. The results indicate optimal performance at α at 0.7 and β at 0.1. Consequently, these values are adopted for all experiments in this paper.

3) Computational Cost: To assess computational efficiency, we compare PhysioNN with other methods on the XDU-SenseTime dataset using four metrics: parameter count, FLOPs, memory usage, and inference time (Table VII). PhysNet achieves the smallest parameter size and memory usage due to its lightweight 3D CNN design. In comparison, our PhysioNN achieves the lowest FLOPs and fastest inference

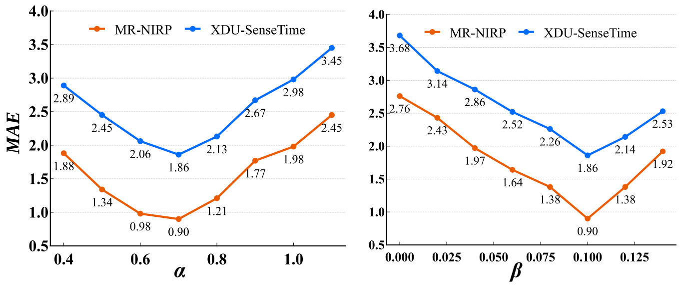

Fig. 6. The impacts of different α and β values on MR-NIRP and XDU-SenseTime datasets.

time owing to its efficient temporal attention modules, making it highly suitable for practical deployment.

TABLE VII
COMPUTATIONAL COST COMPARISON ON XDU-SENSETIME DATASETS. THE BEST RESULTS ARE IN BOLD.

Methods	Para (M)	FLOPs (G)	Memory (M)	Inference Time (ms)
EfficientPhys	1.36	9.67	2.78	97.35
DeepPhys	2.48	9.82	3.61	84.58
TS-CAN	2.17	9.54	2.35	76.49
PhysNet	**0.95**	200.86	**1.47**	119.27
RhythmFormer	3.25	41.31	4.52	215.62
PhysioNN(Ours)	1.09	**6.51**	1.62	**75.49**

IV. CONCLUSIONS

In this paper, the PhysioNN is proposed to effectively extract multi-scale temporal features from rPPG signals, overcoming limitations in HR estimation caused by insufficient physiological representation across different temporal scales in Near-Infrared scenarios. By integrating the BRAM module to capture fine-grained cardiac rhythm patterns (e.g., peak locations and RR intervals) and the PFGR module to ensure waveform fidelity through physiological alignment constraints, our method robustly suppresses waveform distortion caused by noise or motion artifacts while preserving temporal rhythm patterns. Experimental results demonstrate superior performance, with MAE as low as 0.90 bpm on public datasets and strong cross-dataset generalizability. While current work focuses on NIR imaging, future integration of multimodal fusion (e.g., RGB, radar, or WiFi) could leverage complementary information to further enhance accuracy under diverse conditions. Such technology holds significant promise for personalized health assessment, sleep management, and large-scale physiological surveillance in low-resource settings, ultimately contributing to the development of non-contact health monitoring systems.

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